a Participant flow and sample selection

Original merged observations (CHARLS waves 1-5)
n = 96,628

Non-missing residence (city)
n = 96,628

Age >= 45 years
n = 94,240
→ Excluded: 2,388 aged < 45 years

Biomarker waves (2011 and 2015)
n = 37,049
→ Excluded: 57,191 non-biomarker waves

Complete core biomarkers
n = 21,630
→ Excluded: 15,419 incomplete biomarkers

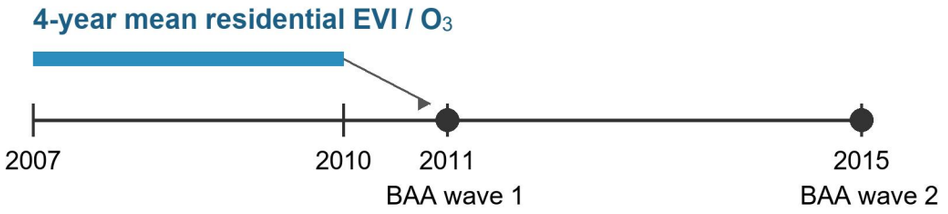
No prevalent major chronic disease at baseline
n = 17,502
→ Excluded: 4,128 prevalent chronic disease at baseline

PhenoAge quality control (fasting, valid ranges, CRP < 10 mg/L)
n = 13,111
→ Excluded: 4,391 failing PhenoAge QC

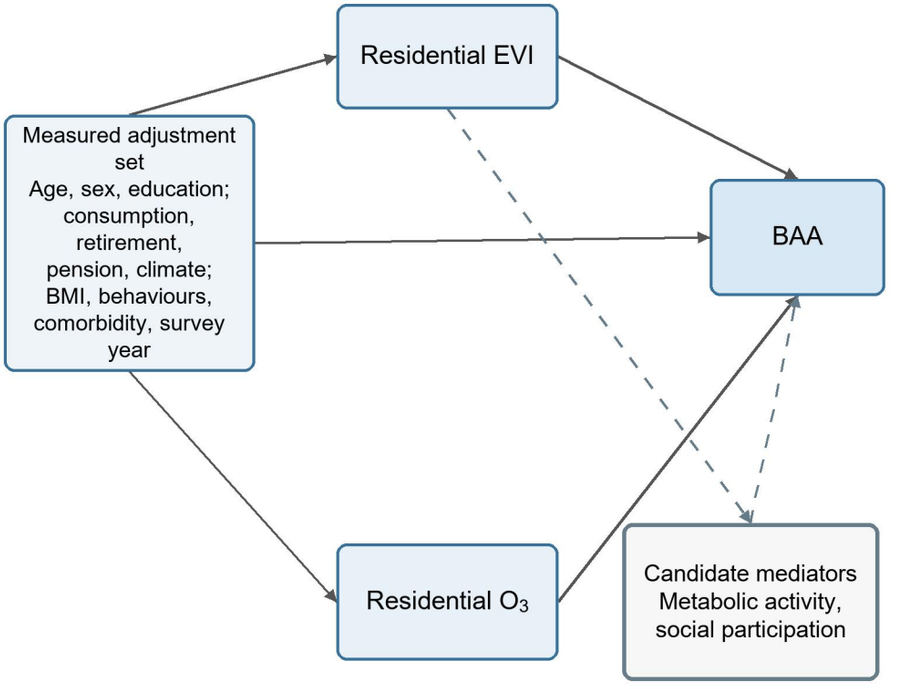
Observed at both biomarker waves
n = 6,568 obs; 3,284 participants
→ Excluded: 6,543 observed at only one biomarker wave

Analytical sample (ages 46-89; complete exposures and covariates)
n = 6,472 obs; 3,281 participants
→ Excluded: 96 outside age range or missing exposures/covariates

b Study timing



c Directed acyclic graph (analysis set)



Solid arrows: working adjustment and exposure-outcome assumptions.
Dashed arrows: exploratory pathways outside the primary models.